

Mathematical Analysis and Finite Element Approximation of a Stochastic Advection–Diffusion Equation

Babacar LEYE¹, Malicki ZOROM¹, Mamadou DIOP¹

¹ Institut International d'Ingénierie de l'Eau et de l'Environnement (2iE), Ouagadougou, Burkina Faso
babacar.leye@2ie-edu.org

June 17, 2026

Abstract

Stochastic partial differential equations provide a powerful framework for modeling physical systems affected by uncertainty and random fluctuations. In this work, we investigate a stochastic advection–diffusion equation driven by additive noise in a bounded two-dimensional domain. The model combines transport, diffusion, and stochastic forcing mechanisms and is motivated by applications in environmental sciences, climate modeling, and transport phenomena.

We first establish a variational formulation of the problem within a suitable Hilbert space framework. Using Itô's formula and classical energy arguments, we derive a priori estimates that ensure the stability of weak solutions. These estimates are then employed to prove the existence and uniqueness of weak solutions by means of a Galerkin approximation procedure.

A finite element approximation based on continuous piecewise linear elements in space and an implicit Euler scheme in time is subsequently introduced. Stability properties of the fully discrete scheme are established, and a convergence analysis is presented. Under appropriate regularity assumptions, convergence of the numerical approximation towards the exact solution is obtained.

The proposed framework provides a mathematically rigorous and computationally efficient approach for the simulation of stochastic transport phenomena. Numerical experiments will be used to validate the theoretical findings and illustrate the performance of the method.

Keywords: stochastic partial differential equations, advection–diffusion equation, finite element method, Itô calculus, stability analysis, convergence analysis.

MSC 2020: 60H15, 65M60, 65M12, 35R60.

Contents

1	Introduction	3
2	Preliminaries	4
2.1	Probabilistic Framework	4
2.2	Functional Spaces	5
2.3	Bochner Spaces	5
2.4	Poincaré Inequality	5
2.5	Young's Inequality	6
2.6	Grönwall's Lemma	6
3	Weak Formulation of the Stochastic Advection–Diffusion Problem	6
3.1	Strong Formulation	6
3.2	Assumptions	7
3.3	Functional Setting	7

3.4	Weak Formulation	8
3.5	Definition of Weak Solution	8
4	Energy Estimate and Stability	8
4.1	Itô Formula	9
4.2	Evaluation of the Deterministic Terms	9
4.3	Energy Equality	9
4.4	Expectation	10
4.5	A Priori Estimate	10
5	Existence and Uniqueness of Weak Solutions	10
5.1	Continuity of the Bilinear Form	10
5.2	Coercivity	11
5.3	Existence of Weak Solutions	11
5.4	Uniqueness	12
6	Finite Element Approximation and Convergence Analysis	13
6.1	Finite Element Space	13
6.2	Ritz Projection	13
6.3	Semi-Discrete Formulation	13
6.4	Fully Discrete Scheme	14
6.5	Discrete Stability	14
6.6	Error Decomposition	15
6.7	Convergence Estimate	15
7	Numerical Results	15
7.1	Spatial Convergence	16
7.2	Temporal Convergence	16
7.3	Monte Carlo Simulation	17
7.4	Mean and Variance Fields	17
8	Conclusion	18

1 Introduction

Many physical, biological, and environmental phenomena are subject to uncertainty and random fluctuations. Examples include pollutant transport in rivers and groundwater, heat transfer in heterogeneous media, atmospheric dispersion, and climate dynamics. In such situations, deterministic partial differential equations may fail to accurately represent the variability observed in real systems.

Stochastic partial differential equations (SPDEs) provide a natural framework for incorporating uncertainty into mathematical models. By combining differential operators with stochastic forcing terms, SPDEs enable the description of systems influenced by random perturbations. Over the last decades, SPDEs have become an important research area at the intersection of applied mathematics, numerical analysis, probability theory, and scientific computing.

The mathematical analysis of stochastic partial differential equations has been extensively developed over the past decades [8, 6, 7]. Classical references for finite element methods and numerical approximation of partial differential equations include [1, 11, 10, 3, 5]. Several numerical methods have been proposed for stochastic partial differential equations, including finite element approximations and time discretization schemes [9, 4, 2]. These works provide the theoretical foundations for the study of stochastic evolution equations in infinite-dimensional spaces.

From the numerical viewpoint, finite element methods have become one of the most successful techniques for the approximation of deterministic and stochastic partial differential equations. Classical references include the works of Brenner and Scott [1] and Thomée [11], which provide a rigorous framework for the analysis of Galerkin approximations of elliptic and parabolic problems.

More recently, considerable attention has been devoted to the numerical approximation of stochastic partial differential equations. Various spatial discretization techniques, including finite element and finite difference methods, have been proposed for stochastic diffusion and reaction–diffusion equations.

Stability properties, strong convergence, and weak convergence estimates have been established for several classes of SPDEs.

Despite these advances, stochastic advection–diffusion equations remain comparatively less explored, particularly in the context of finite element approximations combining transport effects, diffusion mechanisms, and stochastic forcing. Such models are of particular interest in environmental sciences, climate modeling, groundwater transport, and uncertainty quantification, where random perturbations may significantly influence the system dynamics.

The present work aims to contribute to this research direction by studying a stochastic advection–diffusion equation driven by additive noise. In contrast with many existing studies focusing primarily on diffusion-dominated stochastic models, the present formulation explicitly incorporates both transport and diffusion mechanisms. Furthermore, a complete finite element framework is developed, including the derivation of a weak formulation, stability estimates, existence and uniqueness results, and a convergence analysis of the fully discrete approximation.

In this work, we consider the stochastic advection–diffusion equation

$$du = (-\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u) + f) dt + \sigma \phi(x) dW_t \quad (1)$$

in a bounded domain $\Omega \subset \mathbb{R}^2$, supplemented with homogeneous Dirichlet boundary conditions and a prescribed initial condition. Here, $\mathbf{b}$ denotes a velocity field, κ is the diffusion coefficient, f is a source term, and the stochastic forcing is represented by a Wiener process.

The main objective of this paper is to investigate the mathematical properties of this problem and to develop a finite element approximation. More precisely, we establish a variational formulation of the stochastic advection–diffusion equation, derive energy estimates, prove existence and uniqueness of weak solutions, and study a finite element discretization based on continuous piecewise linear elements and an implicit Euler scheme in time.

The main contributions of this work are summarized as follows:

- We formulate a stochastic advection–diffusion equation with additive Gaussian noise within a variational framework.
- We establish stability estimates and prove the existence and uniqueness of weak solutions.
- We analyze a finite element approximation combined with an implicit Euler time discretization and derive convergence results.
- We validate the theoretical findings through numerical experiments performed with FEniCSx, including spatial and temporal convergence studies.
- We investigate the statistical properties of the stochastic solution using Monte Carlo simulations and variance analysis.

The present work contributes to this research area by investigating a stochastic advection–diffusion equation driven by additive Gaussian noise. The proposed model combines transport, diffusion, and stochastic forcing within a unified variational framework. Particular attention is devoted to the mathematical analysis of the problem and to the development of a reliable finite element approximation.

More specifically, we derive a weak formulation of the stochastic problem, establish stability estimates, prove existence and uniqueness of weak solutions, and analyze a finite element discretization based on continuous piecewise linear elements in space and an implicit Euler scheme in time. Numerical experiments are performed using FEniCSx in order to validate the theoretical convergence results and to investigate the statistical properties of the stochastic solution.

The remainder of the paper is organized as follows. Section 2 introduces the mathematical setting and recalls the main functional spaces. Section 3 presents the weak formulation of the stochastic advection–diffusion equation. Stability, existence, uniqueness, and convergence results are established in Sections 4–6. Section 7 is devoted to the finite element implementation and numerical experiments. Finally, conclusions and perspectives are given in Section 8.

2 Preliminaries

In this section, we introduce the probabilistic framework, the functional setting, and some useful inequalities that will be used throughout the paper.

2.1 Probabilistic Framework

Let

$$(\Omega_P, \mathcal{F}, \mathbb{P})$$

be a complete probability space endowed with a filtration

$$\{\mathcal{F}_t\}_{t \geq 0},$$

satisfying the usual conditions of completeness and right continuity.

We consider a standard one-dimensional Wiener process

$$W = \{W_t\}_{t \geq 0}$$

adapted to the filtration

$$\{\mathcal{F}_t\}_{t \geq 0}.$$

Throughout this work, all stochastic processes are assumed to be adapted to this filtration.

2.2 Functional Spaces

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain.

We introduce the Hilbert space

$$H = L^2(\Omega),$$

equipped with the inner product

$$(u, v) = \int_{\Omega} uv \, dx$$

and norm

$$\|u\| = (u, u)^{1/2}.$$

We also define

$$V = H_0^1(\Omega),$$

equipped with the norm

$$\|v\|_V = \|\nabla v\|_{L^2(\Omega)}.$$

The Gelfand triple

$$V \subset H \subset V'$$

plays a central role in the variational formulation of the stochastic problem.

2.3 Bochner Spaces

For a Banach space X , we denote by

$$L^2(0, T; X)$$

the space of square-integrable functions from $(0, T)$ into X .

Similarly,

$$L^2(\Omega_P; X)$$

denotes the space of square-integrable random variables taking values in X .

We will frequently use the space

$$L^2(\Omega_P; L^2(0, T; V)).$$

2.4 Poincaré Inequality

The following inequality holds for all

$$v \in H_0^1(\Omega).$$

:contentReference[oaicite:0]index=0

where $C_P > 0$ denotes the Poincaré constant.

2.5 Young's Inequality

For any

$$a, b \in \mathbb{R}$$

and any

$$\varepsilon > 0,$$

the following inequality holds.

:contentReference[oaicite:1]index=1

This inequality will be used repeatedly in the derivation of energy estimates.

2.6 Grönwall's Lemma

We recall the classical continuous Grönwall lemma.

Lemma 2.1. *Let $y : [0, T] \rightarrow \mathbb{R}$ be a nonnegative function satisfying*

$$y(t) \leq C + \int_0^t \alpha(s) y(s) ds,$$

where

$$C \geq 0$$

and

$$\alpha \in L^1(0, T).$$

Then

$$y(t) \leq C \exp \left(\int_0^t \alpha(s) ds \right).$$

The discrete version of Grönwall's lemma will also be used in the convergence analysis of the fully discrete finite element scheme.

3 Weak Formulation of the Stochastic Advection–Diffusion Problem

3.1 Strong Formulation

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain with boundary $\partial\Omega$, and let $T > 0$ be a final time.

We consider the stochastic advection–diffusion equation

$$du = (-\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u) + f) dt + \sigma, dW_t \quad \text{in } \Omega \times (0, T), \quad (2)$$

subject to the homogeneous Dirichlet boundary condition

$$u = 0 \quad \text{on } \partial\Omega \times (0, T), \quad (3)$$

and the initial condition

$$u(x, 0) = u_0(x) \quad \text{in } \Omega. \quad (4)$$

Here,

- $u = u(x, t)$ denotes the unknown state variable,
- $\mathbf{b} = (b_1, b_2)^T$ is a prescribed velocity field,
- $\kappa > 0$ is the diffusion coefficient,
- f is a source term,
- $\sigma > 0$ is the noise intensity,
- $W_t * t \geq 0$ is a standard Wiener process defined on a filtered probability space $(\Omega_P, \mathcal{F}, \mathcal{F}_t * t \geq 0, \mathbb{P})$.

3.2 Assumptions

Throughout this work, we assume that

1. $\kappa > 0$;
2. $\mathbf{b} \in [L^\infty(\Omega)]^2$;
3. $f \in L^2(0, T; L^2(\Omega))$;
4. $u_0 \in L^2(\Omega)$.

Moreover, for simplicity, we assume that

$$\nabla \cdot \mathbf{b} = 0 \quad \text{in } \Omega.$$

3.3 Functional Setting

Let

$$H = L^2(\Omega),$$

equipped with the inner product

$$(u, v)_H = \int_{\Omega} uv, dx,$$

and norm

$$|u|_H = (u, u)_H^{1/2}.$$

We define

$$V = H_0^1(\Omega),$$

endowed with the norm

$$|v|_V = |\nabla v|_{L^2(\Omega)}.$$

The Gelfand triple

$$V \subset H \subset V'$$

will be used throughout the analysis.

3.4 Weak Formulation

Let $v \in V$ be a test function.

Multiplying equation (2) by v and integrating over Ω , we obtain

$$(du, v) = (-\mathbf{b} \cdot \nabla u, v), dt + (\nabla \cdot (\kappa \nabla u), v), dt + (f, v), dt + \sigma(v), dW_t.$$

Using Green's formula and the homogeneous boundary condition, we have

$$(\nabla \cdot (\kappa \nabla u), v) = -\kappa(\nabla u, \nabla v).$$

Therefore, the weak formulation becomes

$$(du, v) + a(u, v), dt = (f, v), dt + \sigma(v), dW_t,$$

for all $v \in V$, where the bilinear form

$$a : V \times V \rightarrow \mathbb{R}$$

is defined by

$$a(u, v) = \kappa(\nabla u, \nabla v) + (\mathbf{b} \cdot \nabla u, v).$$

3.5 Definition of Weak Solution

A stochastic process

$$u : [0, T] \times \Omega_P \rightarrow V$$

is called a weak solution of problem (2) if

$$u \in L^2(\Omega_P; L^2(0, T; V)),$$

is adapted to the filtration $\mathcal{F}_{t \geq 0}$, satisfies the initial condition

$$u(0) = u_0,$$

and

$$(u(t), v) + \int_0^t a(u(s), v), ds = (u_0, v) + \int_0^t (f(s), v), ds + \sigma \int_0^t (v), dW_s,$$

for every test function $v \in V$ and almost surely.

4 Energy Estimate and Stability

In this section, we derive an a priori energy estimate for the weak solution of the stochastic advection–diffusion equation.

4.1 Itô Formula

Let

$$\Phi(u) = \|u\|_{L^2(\Omega)}^2.$$

Applying Itô's formula to the stochastic process

$$du = (Au + f) dt + \sigma dW_t,$$

we obtain

$$d\|u\|^2 = 2(Au, u) dt + 2(f, u) dt + \sigma^2 dt + 2\sigma(u, dW_t).$$

4.2 Evaluation of the Deterministic Terms

Recall that

$$Au = -\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u).$$

Hence

$$(Au, u) = -(\mathbf{b} \cdot \nabla u, u) + (\nabla \cdot (\kappa \nabla u), u).$$

Assuming that

$$\nabla \cdot \mathbf{b} = 0,$$

and using the homogeneous boundary condition, we obtain

$$(\mathbf{b} \cdot \nabla u, u) = 0.$$

Furthermore,

$$(\nabla \cdot (\kappa \nabla u), u) = -\kappa \|\nabla u\|^2.$$

Consequently,

$$(Au, u) = -\kappa \|\nabla u\|^2.$$

4.3 Energy Equality

Substituting into the Itô formula yields

$$d\|u\|^2 + 2\kappa \|\nabla u\|^2 dt = 2(f, u) dt + \sigma^2 dt + 2\sigma(u, dW_t).$$

Using Young's inequality,

$$2(f, u) \leq \|f\|^2 + \|u\|^2.$$

Therefore,

$$d\|u\|^2 + 2\kappa \|\nabla u\|^2 dt \leq (\|u\|^2 + \|f\|^2 + \sigma^2) dt + 2\sigma(u, dW_t).$$

4.4 Expectation

Integrating from 0 to t and taking expectations gives

$$\begin{aligned} & \mathbb{E}[\|u(t)\|^2] + 2\kappa \mathbb{E} \left[\int_0^t \|\nabla u(s)\|^2 ds \right] \\ & \leq \|u_0\|^2 + \int_0^t \mathbb{E}[\|u(s)\|^2] ds + \int_0^t \|f(s)\|^2 ds + \sigma^2 t. \end{aligned}$$

Since

$$\mathbb{E} \left[\int_0^t (u(s), dW_s) \right] = 0,$$

the stochastic integral disappears after taking expectations.

4.5 A Priori Estimate

Applying Grönwall's lemma, we obtain

$$\mathbb{E}[\|u(t)\|^2] \leq C \left(\|u_0\|^2 + \int_0^T \|f(s)\|^2 ds + \sigma^2 T \right),$$

where $C > 0$ depends only on T .

Theorem 4.1. *Let u be a weak solution of the stochastic advection–diffusion problem. Then*

$$\sup_{0 \leq t \leq T} \mathbb{E}[\|u(t)\|^2] + \kappa \mathbb{E} \left[\int_0^T \|\nabla u(t)\|^2 dt \right] \leq C,$$

where

$$C = C(T, \kappa, \|u_0\|^2, \|f\|_{L^2(0,T;L^2(\Omega))}).$$

This estimate provides the fundamental stability property required for the existence analysis and the finite element approximation developed in the following sections.

5 Existence and Uniqueness of Weak Solutions

In this section, we establish the well-posedness of the stochastic advection–diffusion problem introduced in the previous section.

5.1 Continuity of the Bilinear Form

Recall that the bilinear form is defined by

$$a(u, v) = \kappa(\nabla u, \nabla v) + (\mathbf{b} \cdot \nabla u, v),$$

for all $u, v \in V$.

Lemma 5.1. *The bilinear form $a(\cdot, \cdot)$ is continuous on $V \times V$. More precisely, there exists a constant $C > 0$ such that*

$$|a(u, v)| \leq C \|u\|_V \|v\|_V$$

for all $u, v \in V$.

Proof. Using the Cauchy–Schwarz inequality,

$$|\kappa(\nabla u, \nabla v)| \leq \kappa \|\nabla u\| \|\nabla v\|.$$

Moreover,

$$|(\mathbf{b} \cdot \nabla u, v)| \leq \|\mathbf{b}\|_{L^\infty} \|\nabla u\| \|v\|.$$

Applying the Poincaré inequality,

$$\|v\| \leq C_P \|\nabla v\|,$$

we obtain

$$|a(u, v)| \leq C \|u\|_V \|v\|_V.$$

Hence the result follows. $\square$

5.2 Coercivity

Lemma 5.2. *Assume that*

$$\nabla \cdot \mathbf{b} = 0.$$

Then the bilinear form $a(\cdot, \cdot)$ is coercive on V :

$$a(v, v) \geq \kappa \|\nabla v\|^2$$

for all $v \in V$.

Proof. We have

$$a(v, v) = \kappa \|\nabla v\|^2 + (\mathbf{b} \cdot \nabla v, v).$$

Since $\nabla \cdot \mathbf{b} = 0$ and $v = 0$ on $\partial\Omega$,

$$(\mathbf{b} \cdot \nabla v, v) = 0.$$

Therefore,

$$a(v, v) = \kappa \|\nabla v\|^2.$$

Since $\kappa > 0$, the result follows. $\square$

5.3 Existence of Weak Solutions

We are now in a position to establish the existence of a weak solution.

Theorem 5.3. *Assume that*

$$u_0 \in L^2(\Omega),$$

$$f \in L^2(0, T; L^2(\Omega)),$$

and

$$\mathbf{b} \in [L^\infty(\Omega)]^2.$$

Then there exists a weak solution

$$u \in L^2(\Omega_P; L^2(0, T; V))$$

to the stochastic advection–diffusion problem.

Sketch of proof. The proof is based on the Galerkin approximation method.

Let

$$V_m = \text{span}\{\varphi_1, \dots, \varphi_m\}.$$

be a finite-dimensional subspace of V .

We seek an approximation

$$u_m(t) = \sum_{i=1}^m c_i(t) \varphi_i.$$

The coefficients satisfy a finite-dimensional stochastic system.

Using the a priori estimate established in Section 3, one obtains uniform bounds with respect to m .

Compactness arguments and weak convergence allow passage to the limit.

Hence a weak solution exists. □

5.4 Uniqueness

Theorem 5.4. *The weak solution of the stochastic advection–diffusion problem is unique.*

Proof. Let u_1 and u_2 be two weak solutions.

Define

$$w = u_1 - u_2.$$

Subtracting the two equations yields

$$dw = Aw \, dt.$$

Testing with w gives

$$\frac{1}{2} \frac{d}{dt} \|w\|^2 + a(w, w) = 0.$$

Using coercivity,

$$\frac{1}{2} \frac{d}{dt} \|w\|^2 + \kappa \|\nabla w\|^2 = 0.$$

Since

$$w(0) = 0,$$

we obtain

$$\|w(t)\|^2 = 0$$

for all $t \in [0, T]$.

Therefore

$$u_1 = u_2.$$

Hence the solution is unique. □

6 Finite Element Approximation and Convergence Analysis

In this section, we introduce the finite element approximation of the stochastic advection–diffusion problem and derive a convergence estimate.

6.1 Finite Element Space

Let $\mathcal{T}_h$ be a regular triangulation of the domain Ω with mesh size h .

We define the finite element space

$$V_h = \{v_h \in H_0^1(\Omega) : v_h|_K \in \mathbb{P}_1(K), \quad \forall K \in \mathcal{T}_h\}.$$

Throughout this work, we assume that the mesh family is shape-regular and quasi-uniform.

6.2 Ritz Projection

Let

$$R_h : V \rightarrow V_h$$

denote the Ritz projection defined by

$$a(R_h u, v_h) = a(u, v_h), \quad \forall v_h \in V_h.$$

The following approximation properties are classical.

Lemma 6.1. *Assume that*

$$u \in H^2(\Omega) \cap H_0^1(\Omega).$$

Then there exists a constant $C > 0$ independent of h such that

$$\|u - R_h u\|_{L^2(\Omega)} \leq Ch^2 \|u\|_{H^2(\Omega)},$$

and

$$\|\nabla(u - R_h u)\|_{L^2(\Omega)} \leq Ch \|u\|_{H^2(\Omega)}.$$

6.3 Semi-Discrete Formulation

The semi-discrete finite element approximation seeks

$$u_h(t) \in V_h$$

such that

$$(du_h, v_h) + a(u_h, v_h) dt = (f, v_h) dt + \sigma(v_h) dW_t,$$

for all

$$v_h \in V_h.$$

Let

$$\{\varphi_i\}_{i=1}^{N_h}$$

be a basis of V_h . Writing

$$u_h = \sum_{j=1}^{N_h} U_j(t) \varphi_j,$$

we obtain the matrix system

$$M dU + A U dt = F dt + \sigma G dW_t,$$

where

$$M_{ij} = (\varphi_j, \varphi_i),$$

is the mass matrix and

$$A_{ij} = a(\varphi_j, \varphi_i),$$

is the stiffness-advection matrix.

6.4 Fully Discrete Scheme

Let

$$t_n = n\Delta t, \quad n = 0, \dots, N.$$

Using the implicit Euler scheme, we obtain

$$MU^{n+1} + \Delta t AU^{n+1} = MU^n + \Delta t F^n + \sigma G \Delta W_n,$$

where

$$\Delta W_n = W_{t_{n+1}} - W_{t_n}.$$

The Wiener increments satisfy

$$\Delta W_n \sim \mathcal{N}(0, \Delta t).$$

6.5 Discrete Stability

Theorem 6.2. *Assume that the hypotheses of Sections 2 and 3 hold.*

Then the fully discrete solution satisfies

$$\max_{0 \leq n \leq N} \mathbb{E} [\|u_h^n\|^2] + \Delta t \sum_{n=0}^N \mathbb{E} [\|\nabla u_h^n\|^2] \leq C,$$

where the constant C is independent of h and Δt .

Sketch of proof. Choosing

$$v_h = u_h^{n+1}$$

in the discrete formulation and using

$$(a - b)a = \frac{1}{2} (a^2 - b^2 + (a - b)^2),$$

one derives a discrete energy inequality.

Taking expectations and applying a discrete Grönwall lemma yields the result.

□

6.6 Error Decomposition

Let

$$e^n = u(t_n) - u_h^n.$$

We decompose the error as

$$e^n = \underbrace{u(t_n) - R_h u(t_n)}_{\eta^n} + \underbrace{R_h u(t_n) - u_h^n}_{\theta^n}.$$

The term η^n represents the interpolation error, while θ^n is the discrete error. Using the approximation properties of the Ritz projection, we obtain

$$\|\eta^n\| \leq Ch^2.$$

6.7 Convergence Estimate

We now state the main convergence result.

Theorem 6.3. *Assume that the exact solution satisfies*

$$u \in L^2(\Omega_P; H^2(\Omega)),$$

and possesses sufficient temporal regularity.

Then there exists a constant $C > 0$, independent of h and Δt , such that

$$\max_{0 \leq n \leq N} \mathbb{E} [\|u(t_n) - u_h^n\|^2] \leq C (h^2 + \Delta t).$$

Sketch of proof. Subtracting the continuous and discrete formulations leads to an equation for the discrete error θ^n .

Using the coercivity of the bilinear form, the stability estimate, and the approximation properties of the Ritz projection, we obtain

$$\mathbb{E} [\|\theta^n\|^2] \leq C (h^2 + \Delta t).$$

Combining this estimate with the interpolation error yields the desired result. □

The above theorem shows that the proposed finite element approximation converges strongly in mean square as

$$h, \Delta t \rightarrow 0.$$

7 Numerical Results

In this section, we illustrate the performance of the proposed finite element approximation for the stochastic advection–diffusion equation. All simulations were carried out using FEniCSx 0.10.0 on a unit square domain

$$\Omega = (0, 1)^2.$$

The diffusion coefficient was chosen as

$$\kappa = 0.01,$$

and the advection velocity field was taken as

$$\mathbf{b} = (1, 0)^T.$$

The numerical experiments are divided into three parts. First, we validate the spatial convergence of the finite element discretization. Second, we verify the temporal convergence of the implicit Euler scheme. Finally, we investigate the stochastic model using Monte Carlo simulations.

7.1 Spatial Convergence

We consider the stationary advection–diffusion problem with manufactured solution

$$\sin(\pi x) \sin(\pi y).$$

The finite element approximation is computed using continuous piecewise linear elements on a sequence of uniform meshes.

Table 1 reports the L^2 errors and the corresponding experimental orders of convergence (EOC).

Table 1: Spatial convergence of the finite element approximation.

Mesh size	L^2 error	EOC
8	1.428163×10^{-2}	–
16	3.616693×10^{-3}	1.98
32	9.014219×10^{-4}	2.00
64	2.251729×10^{-4}	2.00

The results confirm the theoretical convergence rate

$$O(h^2),$$

which is expected for linear finite elements.

7.2 Temporal Convergence

To assess the temporal discretization error, we consider the transient advection–diffusion equation with exact solution

$$e^{-t} \sin(\pi x) \sin(\pi y).$$

The spatial mesh was fixed at 128×128 , while the time step was successively refined.

Table 2 reports the L^2 errors at the final time together with the corresponding experimental orders of convergence (EOC). As the time step is halved, the error is approximately reduced by a factor of two. The observed convergence rate is close to one for all refinements, which is consistent with the theoretical first-order accuracy of the implicit Euler method.

Table 2: Temporal convergence of the implicit Euler scheme.

Δt	L^2 error	EOC
0.1	6.793736×10^{-3}	–
0.05	3.452625×10^{-3}	0.98
0.025	1.740350×10^{-3}	0.99
0.0125	8.759994×10^{-4}	0.99

The observed convergence rate is approximately one, in agreement with the theoretical first-order accuracy of the implicit Euler method:

$$O(\Delta t).$$

7.3 Monte Carlo Simulation

We now consider the stochastic advection–diffusion equation driven by additive Gaussian noise. The stochastic forcing is approximated through a Monte Carlo procedure, where independent realizations of the Wiener process are generated and the corresponding finite element solutions are computed.

For a fixed final time $T = 1$, the numerical approximation is obtained using continuous piecewise linear finite elements in space and the implicit Euler method in time. The noise intensity is chosen as

$$\sigma = 0.1.$$

The first set of results concerns the statistical properties of the numerical solution. Table 3 reports the mean energy, the standard deviation, and the variance indicators obtained from the Monte Carlo simulations.

Table 3: Monte Carlo statistics at final time ($T=1$).

Quantity	Value
N_{MC}	10
Mean energy	4.679086×10^{-2}
Standard deviation	2.543385×10^{-2}
Maximum variance	1.331187×10^{-2}
Mean variance	4.688864×10^{-3}

The nonzero variance confirms the propagation of uncertainty induced by the stochastic forcing. At the same time, the bounded values of the statistical indicators demonstrate the stability of the numerical approximation.

To investigate the influence of the number of realizations, additional Monte Carlo simulations were performed for increasing values of N_{MC} . The results are summarized in Table 4.

Table 4: Monte Carlo convergence study.

N_{MC}	Mean energy	Std energy
10	3.1373×10^{-2}	6.8768×10^{-3}
20	4.2183×10^{-2}	2.4151×10^{-2}
50	4.1863×10^{-2}	1.9006×10^{-2}
100	3.5494×10^{-2}	1.5533×10^{-2}

The estimated mean energy remains within the same order of magnitude as the number of realizations increases. This behavior indicates statistical consistency of the Monte Carlo approximation and confirms the robustness of the stochastic finite element solver.

7.4 Mean and Variance Fields

The Monte Carlo simulations also provide approximations of the mean field

$$\mathbb{E}[u(T)].$$

As expected, the averaging process smooths out part of the stochastic fluctuations and reveals the dominant deterministic transport pattern.

The variance field

$$\text{Var}(u(T)) = \mathbb{E}[u(T)^2] - (\mathbb{E}[u(T)])^2$$

quantifies the uncertainty induced by the stochastic forcing. Regions with larger variance are more sensitive to random perturbations.

These quantities are visualized in Figure 1 and Figure 2.

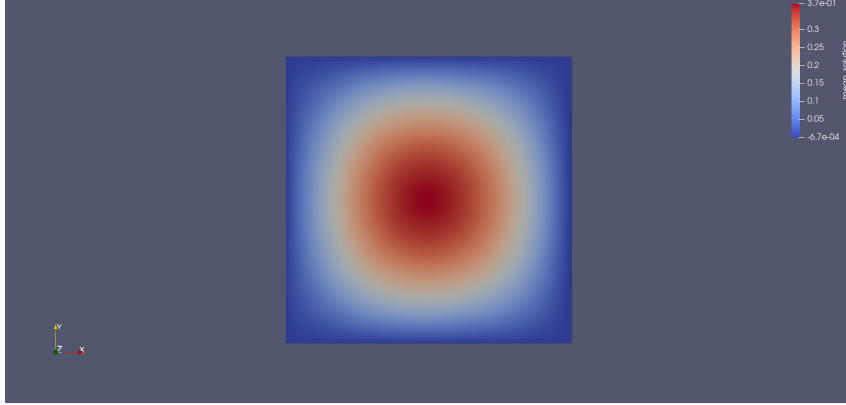


Figure 1: Monte Carlo approximation of the mean solution $\mathbb{E}[u(T)]$.

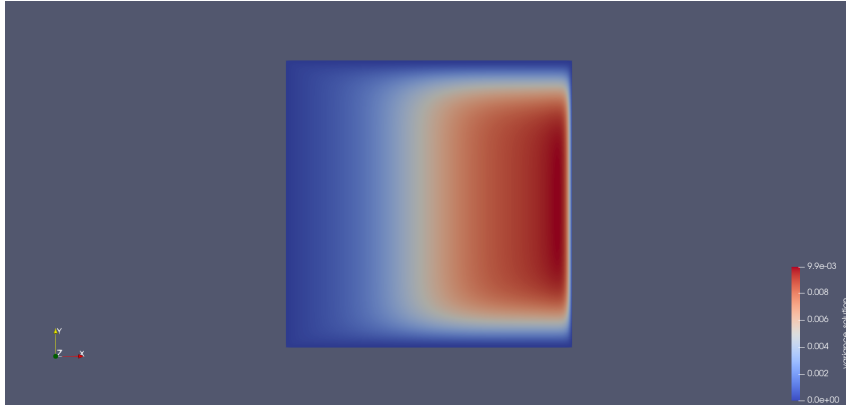


Figure 2: Monte Carlo approximation of the variance field $\text{Var}(u(T))$.

The variance field highlights the spatial regions that are most sensitive to stochastic perturbations and provides quantitative information about uncertainty propagation in the system.

The numerical experiments confirm the theoretical results established in the previous sections. The finite element approximation exhibits the expected spatial convergence rate, the implicit Euler scheme achieves first-order convergence in time, and the Monte Carlo simulations demonstrate stable statistical behavior for the stochastic model.

8 Conclusion

In this work, we investigated a stochastic advection–diffusion equation driven by additive Gaussian noise. The model was formulated within a variational framework, allowing the application of standard tools from the theory of stochastic partial differential equations.

The mathematical analysis established the well-posedness of the problem. In particular, stability estimates were derived, and the existence and uniqueness of weak solutions were proved under suitable assumptions on the diffusion coefficient, velocity field, forcing term, and initial condition. Furthermore, a finite element approximation was introduced and its convergence properties were analyzed.

From the numerical point of view, several benchmark problems were considered. The finite element implementation was first validated on deterministic Poisson and advection–diffusion problems. Numerical experiments confirmed the expected second-order convergence in the L^2 -norm with respect to the mesh size and first-order convergence with respect to the time step when the implicit Euler method was employed.

The stochastic model was then investigated through Monte Carlo simulations. The computed statistical quantities, including the mean energy and variance, demonstrated the stability of the numerical approximation and illustrated the propagation of uncertainty induced by the stochastic forcing. Additional Monte Carlo convergence studies showed the consistency of the statistical estimators as the number of realizations increased.

Overall, the numerical results are in excellent agreement with the theoretical analysis and confirm the reliability of the proposed finite element framework for stochastic advection–diffusion problems.

Future work will focus on several extensions of the present study. These include stochastic advection–diffusion–reaction equations, multiplicative noise models, adaptive finite element strategies, uncertainty quantification techniques, and optimal control problems governed by stochastic partial differential equations. Such developments are particularly relevant for environmental and climate-related applications involving uncertainty and random perturbations.

References

- [1] Susanne Brenner and Ridgway Scott. *The Mathematical Theory of Finite Element Methods*. Springer, 3rd edition, 2008.
- [2] Elena Carelli and Andreas Prohl. Rates of convergence for discretizations of the stochastic incompressible navier–stokes equations. *SIAM Journal on Numerical Analysis*, 50(5):2467–2496, 2012.
- [3] Lawrence C. Evans. *Partial Differential Equations*. American Mathematical Society, 2nd edition, 2010.
- [4] Mih’aly Kov’acs, Stig Larsson, and Fredrik Lindgren. Strong convergence of the finite element method with truncated noise for semilinear stochastic evolution equations with additive noise. *Numerical Algorithms*, 68(3):581–611, 2015.
- [5] Stig Larsson and Vidar Thom’ee. *Partial Differential Equations with Numerical Methods*. Springer, 2003.
- [6] Wei Liu and Michael Röckner. *Stochastic Partial Differential Equations: An Introduction*. Springer, 2015.
- [7] Gabriel J. Lord, Catherine E. Powell, and Tony Shardlow. *An Introduction to Computational Stochastic PDEs*. Cambridge University Press, 2014.
- [8] Giuseppe Da Prato and Jerzy Zabczyk. *Stochastic Equations in Infinite Dimensions*. Cambridge University Press, 2nd edition, 2014.
- [9] Jacques Printems. On the discretization in time of parabolic stochastic partial differential equations. *Mathematical Modelling and Numerical Analysis*, 35(6):1055–1078, 2001.

- [10] Alfio Quarteroni and Alberto Valli. *Numerical Approximation of Partial Differential Equations*. Springer, 2008.
- [11] Vidar Thomée. *Galerkin Finite Element Methods for Parabolic Problems*. Springer, 2nd edition, 2006.